🎬 Shot Designer

Top-down blocking & camera planning in the browser — User Manual

1. What is it?

Shot Designer is a browser app for planning scene blocking: an orthographic top-down view where you place characters, props and cameras over your own floor plan, animate their movement with keyframes, cut between cameras as in a live edit, and export the result as a video or an image sequence.

2. Getting started

- Serve the project folder with any static server (e.g. `python -m http.server 8741`) and open `http://localhost:8741/`. Opening `index.html` directly also works.
- The app opens with a small demo scene. Press 🗑 New to start from scratch.
- Language: use the ENG | SPA toggle at the top-left. The choice is remembered.

3. Interface overview

- **Top bar** — add elements, load a backdrop, global toggles (Names, Paths), Render, Save / Open / New.
- **Camera switcher** (red bar) — one button per camera; shows which one is LIVE. Click (or press `1`–`9`) to cut.
- **Canvas** (center) — the stage, seen from above. Scroll wheel zooms, dragging empty space (or right-drag) pans.
- **Inspector** (right) — properties of the selected element. With nothing selected it shows the shortcut cheat-sheet.
- **Timeline** (bottom) — playback controls, duration, one row per element with its keyframes, and the CUTS track.

4. Adding elements

Characters

- ♂ Male / ♀ Female add a character at the view center. Genders are visually distinct: females have long hair with a ponytail and a flared dress; males have broad shoulders and cropped hair. Labels carry a ♀/♂ symbol.
- In the inspector: rename, switch gender, pick one of 6 **skin tones**, choose a **clothes color** (palette or free color picker) and adjust **size**.
- Reading facing at a glance: **eyes and nose = front**, hair/ponytail = back. The blue rotation handle also points forward.

Cameras

- 🎥 Camera adds a numbered camera (C1, C2...) with a field-of-view cone. Adjust **Angle (FOV)** and **Range** in the inspector.
- **Handheld** slider (0–100%): adds organic hand-held shake to the cone/body. Deterministic — replays identically every time.
- **POV**: 👁 POV adds a camera bound to the selected character; it follows the character's position and facing automatically ("Point of view" in the inspector can bind/unbind any camera).

Props

- ▢ Prop adds a box or circle (shape, width, height, color) to mark furniture or zones.

Backdrop

- 🖼 Backdrop loads your own image (floor plan, set sketch). Scale and opacity sliders appear next to the button; ✕ removes it. The image is stored inside the scene file.

5. Animating with keyframes

> **The basic workflow:** keep `● AUTO-KEY` on (it is by default). Move the playhead to second 5, drag the character where it should be — a keyframe is recorded automatically. Move the playhead to second 10, drag again. Press `Space` to watch it.

- **Duration**: set the scene length in the timeline bar (1–30 s max).
- **Timeline editing**: each diamond is a keyframe. Drag it to move it in time; right-click deletes it. The "+ Keyframe at Xs" button in the inspector adds one at the playhead.
- **AUTO-KEY off**: dragging moves the whole path (all keyframes shift together).
- **Rotation**: drag the blue dot next to the selected element. Hold `Shift` for 15° snapping.
- **Easing**: select a keyframe diamond, then choose Linear / Ease in / Ease out / In-Out in the inspector — it applies to the segment leaving that keyframe. With no keyframe selected, the buttons apply to all segments. Eased keyframes are drawn round.
- **Separate pos/rot channels**: "Pos/rot keyframes → Separate" makes moving record only position and rotating only rotation, as independent channels (cyan diamonds = position, green = rotation).

6. Paths and curves

- Each moving element shows its **path**: a dashed line with direction arrows and a time label on every keyframe dot.
- **Bend a segment**: select the element and drag the **orange square** in the middle of a segment — it becomes a bezier curve. The character walks along the curve and the arrows follow it.
- **Straighten**: right-click the orange square (one segment) or use "Straighten path" in the inspector (all segments). Holding `Shift` while dragging also snaps straight.
- **S / snake curves**: double-click anywhere on the path to insert a keyframe at that point *without changing the shape*. Each half then has its own orange handle — bend them opposite ways.
- **Automatic facing**: characters default to "Follow path (auto)" — they look where they are heading, turning smoothly along curves. Grabbing the rotation handle switches them to Manual; switch back any time in "Facing / rotation". Free cameras can also opt into auto.
- Visibility: the ➜ button per row toggles one path; the top-bar `➜ Paths` toggle hides all of them at once.

7. Live camera cutting

- The switcher bar lists every camera. The **LIVE** one is highlighted red on stage (red cone) and in the bar.
- Press `1`–`9` or click a camera button to **cut** to it at the current time — do it during playback to edit "live". Each cut lands on the red **CUTS** track in the timeline: drag markers to retime, right-click to delete.
- During playback the LIVE camera updates in real time following your cuts, like a multicam program feed.

8. Showing / hiding things

- Every element has an eye (👁) in its timeline row and a "Visible" button in the inspector.
- **Labels**: top-bar `🏷 Names` hides all names. Per element, the "Label" section lets you hide its name, remove or recolor the background box, change the text color and the size (8–30).

9. Rendering

- `🎬 Render ▾` opens a menu with two outputs. What you see is what you get: renders are clean (no grid, no selection rings), but names, paths and camera cones are included *if* their toggles are on — turn them off first for a clean video, or leave them for a technical preview.

- **Video (MP4, max 30 MB)**: plays the whole scene once while recording at 30 fps. The bitrate is budgeted automatically so the file never exceeds ~30 MB (quality-capped at 10 Mbps). Chrome/Edge download a native MP4; browsers that cannot mux MP4 get a WebM with a notice.
- **Images (1 every 3 s, ZIP)**: captures a PNG at 0s, 3s, 6s… plus the final frame, at full canvas resolution, and downloads them as a single `shotdesigner-frames.zip`.

10. Saving, opening, undo

- 💾 Save (or `Ctrl`+`S`) downloads the whole scene as a JSON file — characters, keyframes, curves, cuts, labels and the backdrop image included. 📂 Open loads it back.
- **Autosave**: the scene is also saved continuously in the browser (localStorage) — closing the tab loses nothing on the same machine/browser.
- **Undo / redo**: `Ctrl`+`Z` / `Ctrl`+`Shift`+`Z` (or `Ctrl`+`Y`), up to 60 steps. Each drag or discrete action is one step.

11. Keyboard shortcuts

Key	Action
`Space`	Play / pause
`1` – `9`	Cut to that camera at the current time (live editing)
`Ctrl`+`Z` / `Ctrl`+`Shift`+`Z` / `Ctrl`+`Y`	Undo / redo
`Ctrl`+`S`	Save (downloads the scene JSON)
`←` / `→`	Step the playhead 0.1 s (with `Shift`: 1 s)
`Del` / `Backspace`	Delete the selected keyframe, or the selected element
`Esc`	Deselect
`Shift` while rotating	Snap to 15°
`Shift` while bending a path	Snap segment straight
Mouse wheel	Zoom (centered on cursor)
Drag empty space / right-drag	Pan the view
Double-click on a path	Insert a keyframe at that point (shape preserved)
Right-click orange square	Straighten that segment
Right-click keyframe / cut marker	Delete it

12. Tips

- Draw your floor plan in any tool, export it as PNG/JPG and load it as the backdrop — then block over it.
- Plan the movement first with paths visible; when presenting, hide Names and Paths and render the clean video.
- Use a POV camera plus a cut to it to preview what a character sees at a given moment.
- Keep an eye on the LIVE bar while playing — rehearse your cuts by pressing number keys in real time, then fine-tune the markers on the CUTS track.